University of Cape Town
Department of Computer Science

Computer Science CSC3003S
November Examination, 2018

Read through the entire examination paper before you start writing so that you can answer the easy questions first. The relative importance of a question is indicated by the marks allocated to it. Please write legibly.

This paper consists of **Four Sections:**

Section A: *Advanced Software Design 1*

Section B: *Advanced Software Design 2*

Section C: *Theory of Algorithms*

Section D: *Theory of Computation*

Total Marks: 90	Approximate marks per question are shown in brackets.
Time: 3 hours + 10 minutes reading	The use of calculators is permitted.

Please write your answers in FOUR separate books. Use one book for Advanced Software Design 1, one book for Advanced Software Design 2, and one book each for Theory of Algorithms and Theory of Computation. **Label each of the four books as ASD1, ASD2, ToA and ToC**.

Your attention is drawn to the instructions on your answer book on the front and the back. In particular, "**EVERY CANDIDATE MUST enter the book number and the number of each question answered (in the order in which it has been answered)**" on the table provided on the front cover of the first book.

Please **do this for the front cover of the different books used for each section.**

This is needed so that we can allocate marks and not miss questions. **A list of sub-questions is *not* required.**

Section A. [21]

Question 1. **[6]**

a) Define Software Engineering in one sentence [2]

b) What is version/source control? [2]

c) Describe an advantage of using a distributed source control such as git? [1]

d) Which git command should be used to create a new git repository? [1]

Question 2. **[9]**

a) You have been asked to plan a project to build a cryptocurrency exchange app to allow users to easily buy their favourite cryptocurrenices. Draw the network analysis diagram based on the following information: [7]

#	Activity	Length	Dependencies
1	Architecture	30	
2	Database	40	1
3	Market Making	20	2; 4
4	Coin listing	20	8
5	Deposit	10	4
6	Withdrawal	10	4
8	Trade platform	40	
9	Troll box	5	
10	Launch	20	3; 5; 6; 11; 9
11	Legal	100	

b) Highlight the critical path [2]

Question 3. **[6]**

Agile Manifesto

We are uncovering better ways of developing software by doing it and helping others do it. Through this work we have come to value:

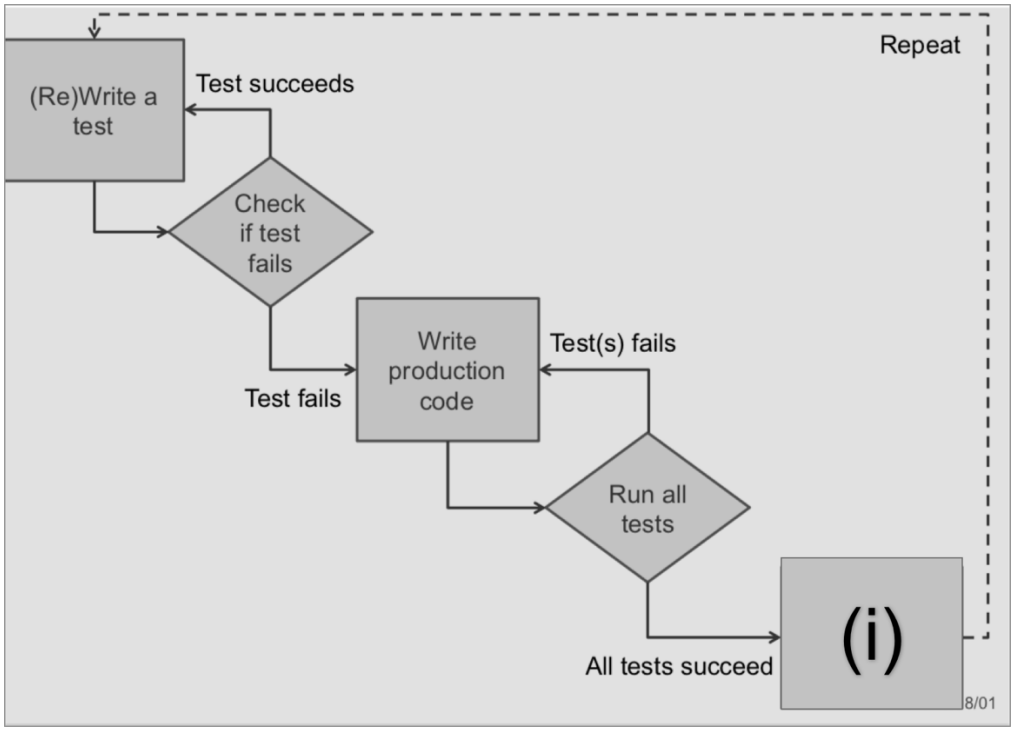
- **Individuals and interactions** over **processes and tools**
- ? over ?
- ? over ?
- ? over ?

That is, while there is value in the items on the right, we value the items on the left more.

a) Read the agile manifesto above. Provide any one of the missing lines. [2]

b) One of the Extreme Programming principles is Pair programming. Developers work in pairs, sitting together to write each line of code on one machine. Discuss two advantages of using Extreme Programming. [2]

c) Testing is one of the most important parts of any software project. Examine the diagram below.



i) Fill in the missing text indicated by (i) in the above diagram. [1]

ii) Name an advantage of (i) [1]

Section B. [23]

Question 4. **[5]**

a) One of the main goals of software architecture is abstraction. Explain the meaning of this term. [2]

b) Major decisions are made during the Architectural Design phase of a project. Name two important decisions that are made. [2]

c) How does using a design pattern like layered architecture assist with work allocation? [1]

Question 5. **[10]**

a) You have recently been appointed as the software architect of a new software system for a business which has to process large volumes of paper documents, which it receives every day. Received documents need to go through the following process. The documents have to be scanned, scanned document images need to be enhanced, document information needs to be captured by data capturers from the document images, and then finally the document images along with their captured information need to be stored in a database. Briefly explain what the most sensible underlying architectural design pattern to use is, why you chose it, and use a diagram to illustrate the actual system. [8]

b) Briefly explain how this architectural model you have selected above can be used to process documents faster. [2]

Question 6. **[8]**

a) Many file sharing applications on the internet use an alternative architectural design pattern to the one used more commonly by services like email and the world wide web. What is this alternative architectural design pattern called? [1]

b) What purpose do Architectural Models serve after the system has been developed? [2]

c) Briefly describe how batch processing systems work in terms of the standard components and levels of user interaction. [2]

d) A word processor application is a typical example of what kind of generic application architecture? [1]

e) Briefly explain the GRASP Information Expert design pattern. [1]

f) Although open source is free, the free software movement, which precedes open source, has a particular understanding of what the term “free” means. Can you clarify what that understanding is. [1]

Section C. [23]

Question 7. **[13]**

a) Explain **any five** (5) of the following problem solving techniques (about 1 line each):

Brute force, Decrease and conquer, Divide and conquer, Dynamic programming, Greedy, Space-time-tradeoff, Transform-and-conquer [5]

b) Mult(a,n) computes a * n and would be computed as below using brute force:

Mult(a,n) = a + a + a + …. + a (n times)

Show how it would be computed using each technique below: [3]

1. Decrease by a constant
2. Decrease by a constant factor
3. Divide and conquer

c) **Which two** statements below are True? [2]

1. $O(n + \log n) = O(n)$
2. $\Theta(\log^2 n) = \Theta(\log n)$
3. if $x \in O(n \log n)$ then $x \in \Omega(n)$
4. if $x \in \Theta(2^n)$ then $x \in \Omega(n)$

d) What would be the values of **a, b and d** if we used the Master Theorem to find the efficiency of the following function in terms of the number of additions? **Just give the 3 values, showing which is which**.

waat (x) { if x = 0 return a[0]; a[x] = waat(2x/5) + waat(2x/5); return a[x] }

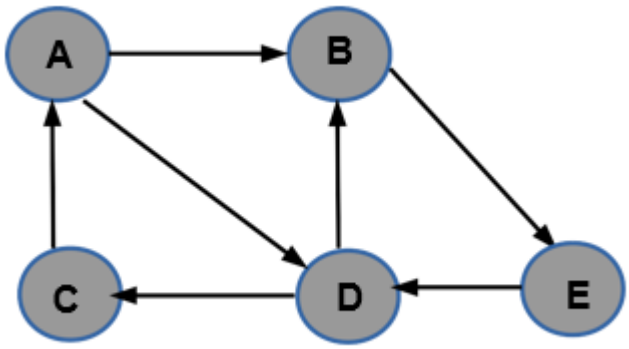
Master Theorem: If $T(n) = aT(n/b) + f(n)$ then $T(n)$ is $O(n^d)$ if $a < b^d$

$T(n)$ is $O(n^d \log n)$ if $a = b^d$

$T(n)$ is $O(n^{\log_b a})$ if $a > b^d$ [3]

Question 8. **[10]**

a) Consider the graph below and answer the questions that follow.



1. Show the initial array A(1) that would be used by Warshall's algorithm that finds the transitive closure of a graph. You may leave elements that are zero blank, and only show where the 1's are. [2]
2. Which element(s) of this array would change from zero to 1 in A(2) ? Just give their index(es), e.g. A[8,9]. Assume nodes are processed in alphabetical order. [1]
3. What is the complexity of Warshall's algorithm and why? [1]
4. Floyd's all pairs shortest paths algorithm is based on Warshall's idea, but it returns paths, e.g. X→W→Y→Z, not just path lengths. How does it adapt Warshall's algorithm so as to be able to return paths as well? State the solution briefly in one line, don't give details. [1]
5. Dijkstra's algorithm differs from Floyd's algorithm because it finds shortest paths to all other nodes from a single start-node only. Give one other difference between these 2 algorithms' functionality (**what** they can do, **not how** they do it). [1]
6. "One of the many uses of graphs in problem solving is for transform-and-conquer state-space search". Either explain this statement or give an example of this. [1]

b) Chose EITHER ONE of the problems below and then give **two different algorithms** (in words or pseudocode) for that same problem. [3]

EITHER

1. There is a Lucky Number competition where you pay R10 to guess a number between 1 and 100 000 000. Whoever guesses the Lucky Number wins a million Rand. Every guess wins some money, but the amount won depends how close it is to the Lucky Number. The further a guess is from the Lucky Number the less you win; the closer a guess is to the Lucky Number, the more you win. How do you go about finding the Lucky Number in as few guesses as possible?

OR (PLEASE TURN OVER)

2. Given a triangle of digits as shown below, find the lowest-cost path from the top to any digit on the bottom. The cost of a path is the sum of all the digits on that path. To get from any digit to the next in a path you can only go to the number directly below and 1 position to its left, or else to the number directly below and 1 position to its right. For the example below the lowest cost is 12 (path 1 3 8 0). The triangle will have between 1 and 10 000 rows (i.e. a path will be of length between 1 and 10 000).

```
   1
  2 3
 4 8 6
7 8 0 1
```

Section D. [23]

Question 9. **[5]**

Let L_n be the language containing all words of the form a^k where k is a multiple of n, and $n \geq 0$ (k is a multiple of n if there is a natural number x such that $k = x*n$). For example, the language $L_2 = \{\varepsilon, aa, aaaa, aaaaaa, \ldots\}$. Give a proof outline that for every $n \geq 0$, B_n is regular.

Question 10. **[4]**

Prove that the following language is not regular: $L = \{ w \mid w \in \{a,b,c,d,e\}^* \text{ and } w = w^r\}$. So L is a palindrome made up of the letters a, b, c, d, and e.

Question 11. **[5]**

Draw a PDA that accepts the following language: $L = \{ a^m b^n \mid m \geq n \text{ and } n \geq 0 \}$.

Question 12. **[4]**

Prove that the context-free languages are closed under union.

Question 13. **[5]**

Prove that if a language is Turing-recognisable, and its complement is Turing-recognisable, then the language is decidable.

-END-